

AI in Material Science: AI-Powered Cosmetic Ingredient Analysis and Safety Recommendation System Using OCR and Explainable AI

1. BASE PAPER

Base paper: Sayallar, Ç., & Sayar, A. (2025), “Harmful ingredient detection from cosmetic products using optical character recognition and bi-LSTM model.”

Why this is the best base paper: It is the closest recent peer-reviewed study to the proposed research because it starts from a photograph of a cosmetic label, uses OCR to extract text, applies a deep-learning information-extraction model to identify ingredients, and compares the detected ingredients with a harmful-ingredient database. The proposed research can extend this pipeline by adding explainable AI, personalized safety scoring, skin-profile-aware recommendations, uncertainty/confidence, and evidence-backed explanations.

Proposed research gap: Existing work largely focuses on detecting/extracting ingredients or predicting a particular safety endpoint. A stronger integrated system can connect (1) label-image OCR, (2) ingredient normalization/entity matching, (3) multi-dimensional safety prediction, (4) Explainable AI such as SHAP/feature attribution, and (5) personalized recommendation with a clear evidence trail. This makes the system more useful for consumers while keeping the recommendation transparent and safety-oriented.

2. LITERATURE SURVEY

The following survey prioritizes 2024–2026 work and covers the five technical pillars of the proposed study: OCR/ingredient extraction, cosmetic recommendation, AI-based formulation, skin/chemical safety prediction, and explainable AI.

Reference	Contribution	Methodology	Data set	Results	Key Findings
[1] Natsir & Mustafa (2026)	OCR-LSTM analysis for hazardous cosmetic ingredients and skin-type suitability.	OCR → preprocessing/normalization → LSTM classification.	Cosmetic ingredient data; BPOM hazardous-ingredient list; skin-type categories.	Training accuracy 97.59%; validation accuracy 96.68%.	Directly supports OCR, safety classification, and personalization.
[2] Suryaprasad et al. (2026)	Uses OCR and RAG to produce evidence-grounded skin-care recommendations.	Label OCR + skin profile → hierarchical RAG → evidence retrieval → recommendation.	850 query responses and retrieved dermatology evidence.	F1 0.889; grounding accuracy 97.3%; latency 3.2 s.	Supports evidence-grounded recommendation and explainability.
[3] Ganga Sri et al. (2026)	Integrates OCR, ingredient matching, risk scoring, and skin profiling.	OCR → fuzzy matching → multidimensional risk scoring → skin profiling.	More than 500 organic cosmetic compositions.	Hazardous-ingredient accuracy 87.3%; skin-type accuracy 92.1%.	Very close to the proposed end-to-end architecture.
[4] Qayyum et al. (2026)	Reviews AI, QSAR, and in-silico strategies for skincare safety prediction.	Critical review of computational toxicology and AI-based safety methods.	Published skincare-safety and computational-toxicology literature.	Highlights interpretability, robustness, overfitting, and regulation challenges.	Supports selection of safety endpoints and XAI.
[5] Ambe et al. (2026)	Introduces an explainable model for skin-	XGBoost + molecular descriptors/assays + structural alerts + SHAP.	154 internal and 38 external substances.	80% internal and 65% external predictions within	Strong recent foundation for SHAP-based XAI.

	sensitization risk assessment.			five-fold measured range.	
[6] Lee et al. (2026)	Predicts skin sensitization of consumer-product ingredients with mechanistic AI.	GCNs predict key events; biological weighting follows an adverse-outcome pathway.	Consumer-product/fragrance ingredient screening set of 3,415 compounds.	AUC 0.81–0.90 across key events; potential sensitizers identified.	Supports mechanistically interpretable ingredient safety screening.
[7] Sayallar & Sayar (2025)	Detects potentially harmful cosmetic ingredients directly from label images.	Image → OCR → bi-LSTM extraction → harmful-ingredient matching.	Cosmetic product labels and a harmful-ingredient database.	Reports high-accuracy ingredient identification and mobile risk workflow.	Best base paper; proposed work adds XAI and recommendations.
[8] Liu et al. (2025)	Makes cosmetic recommendation safety-aware by modelling sensitization risk.	Three-branch hybrid model combines statistical, text, and allergy features.	Cosmetic recommendation data with ingredient/allergy features.	Recommendation accuracy above 0.90; three-level safety classification.	Shows safety should be part of recommendation.
[9] Di Guardo et al. (2025)	Reviews AI for cosmetic formulation, safety, tolerability, and regulation.	Review of predictive modelling, toxicology, semi-supervised and RL methods.	PubMed/Scopus literature on AI and cosmetic formulation.	AI can accelerate safety screening; data quality and interpretability remain issues.	Provides the broad AI-in-cosmetics foundation.
[10] Liu et al. (2025)	Uses explainable ML to predict eye/skin irritation and corrosivity.	ML/DL models with explanations at dataset, molecular, and atomic levels.	3,316 eye-irritation and 3,080 skin-irritation data points.	External balanced accuracy 73.0% eye and 75.1% skin.	Strong reference for understandable safety explanations.
[11] Lechuga et al. (2025)	Predicts surfactant-induced skin irritation from formulation properties.	Regression and conditional inference tree using HLB, concentration, ionic character.	20 commercial surfactants and binary mixtures.	Regression explained 80.1% variability; classification accuracy 72%.	Shows formulation properties can support transparent safety scoring.
[12] Xie et al. (2025)	Combines molecular structure and toxicological knowledge for toxicity prediction.	Knowledge graph + GCN/GAT/R-GCN/HGT/GPS models.	Tox21 plus PubChem, Reactome, ChEMBL, and ComptoxAI knowledge.	Knowledge-enriched GNNs outperformed structure-only baselines.	Supports a knowledge-grounded ingredient safety database.
[13] Zhang et al. (2025)	Uses transfer learning and attention-enhanced GNNs for toxicity prediction.	GNN pretrained on baseline toxicity/lipophilicity; fine-tuned for target toxicity; SHAP.	Chemical toxicity data with baseline toxicity/logP information.	Competitive performance with meaningful structural alerts.	Useful when cosmetic-specific labelled data are limited.
[14] Lee et al. (2024)	Uses deep learning for ingredient-based skin-care recommendation.	Transformer encodes ingredient sequences; skin-analysis models characterize conditions.	Cosmetic ingredient sequences and facial skin-condition data.	Effective efficacy prediction and personalized recommendations.	Supports the personalized recommendation module.
[15] Xin et al. (2024)	Reviews AI/ML for critical cosmetic materials and formulation design.	Review covering surfactants, polymers, fragrances, preservatives, and hydrogels.	Recent literature on cosmetic material/formulation AI.	AI can reduce trial-and-error and support allergen screening.	Connects the project to AI in material science.
[16] Grech et al. (2024)	Reviews AI applications in cosmetology and responsible	Literature review of AI/ML for skin analysis, recommendation, safety, and ethics.	Published AI/cosmetology literature.	Emphasizes accuracy, privacy, ethics, and bias.	Supports responsible user-facing AI design.

	personalized care.				
[17] Huang et al. (2024)	Develops explainable ML models for skin corrosion and irritation prediction.	Five ML algorithms with six molecular representations and explainability analysis.	731 corrosion, 1,283 irritation, and 1,205 negative samples.	Provides predictive models and interpretable structural alerts.	Strong foundation for ingredient-level XAI.
[18] Wang et al. (2024)	Combines ML/QSAR evidence for skin-sensitization prediction.	Multiple QSAR models + Dempster-Shafer evidence combination.	GPMT, LLNA, DPRA, KeratinoSens, and h-CLAT datasets.	Test-set correct classification ranged 0.688–0.764.	Supports multi-source safety evidence fusion.
[19] Qiao et al. (2024)	Predicts skin-sensitization potential specifically for cosmetic compounds.	HaCaT/THP-1 co-culture generates signals used by ML models.	Cosmetic-compound sensitization data from co-culture assays.	Demonstrates ML support for allergen/sensitization screening.	Provides cosmetic-specific safety modelling.
[20] Kong et al. (2024)	Detects cosmetic sensitizers while estimating prediction reliability.	Graph autoencoder + Mondrian conformal prediction.	589 sensitizers and 831 non-sensitizers.	Produces uncertainty-aware confidence for predictions.	Useful for confidence-aware safety recommendations.

3. REFERENCES

- [1] Natsir, M. S., & Mustafa, M. S. (2026). An OCR–LSTM-based framework for hazardous cosmetic ingredient detection and skin-type classification using ingredient analysis. *Indonesian Journal of Enterprise Architecture*, 3(2), 12–22. <https://doi.org/10.66314/ijea.v3i2.433>
- [2] Suryaprasad, L., Joshi, V., Eshwari, Haneen, A., & Kendagannaswamy, M. S. (2026). Chemical compounds recommendation engine based on skin types using retrieval-augmented generation and optical character recognition. *ICTACT Journal on Data Science and Machine Learning*, 7(3), 1091–1096. <https://doi.org/10.21917/ijdsml.2026.0200>
- [3] Ganga Sri, S., Mohamed Faisal, J., Mohamed Suhail, I., & Sarmathi, S. (2026). Glow Shield: Smart cosmetic ingredient analyzer for skin risk prediction. In *Current Research in Chemical, Biological, Healthcare and Data Science*. Bhumi Publishing.
- [4] Qayyum, H., Mujtaba, G., Abidin, M. B. Z., Zaini, D. B., Ahmad, S. I. B., & Hai, A. (2026). Recent advances in computational modelling, artificial intelligence, and in silico strategies for predicting adverse effects in skincare safety: A critical review. *European Journal of Medicinal Chemistry Reports*, 17, 100338. <https://doi.org/10.1016/j.ejmcr.2026.100338>
- [5] Ambe, K., Kinoshita, K., Tokunaga, J., Satake, R., Murasaki, W., Tohkin, M., Yamada, T., & Ashikaga, T. (2026). ExSERA: The explainable machine learning model for skin sensitization risk assessment. *Regulatory Toxicology and Pharmacology*, 169, 106100. <https://doi.org/10.1016/j.yrtph.2026.106100>
- [6] Lee, S., Kim, H., Kim, S., & Seo, M. (2026). SkinCast: An AI-driven mechanistically interpretable model for predicting skin sensitization of environmentally released consumer product ingredients. *Environment International*, 210, 110209. <https://doi.org/10.1016/j.envint.2026.110209>
- [7] Sayallar, Ç., & Sayar, A. (2025). Harmful ingredient detection from cosmetic products using optical character recognition and bi-LSTM model. *Signal, Image and Video Processing*, 19(4). <https://doi.org/10.1007/s11760-025-03923-0>
- [8] Liu, W., Yan, C., & Shao, Y. (2025). Safety-aware cosmetic recommendation system via three-branch hybrid modeling: Highlighting sensitization among multidimensional features. *Proceedings of the 2025 8th International Conference on Computer Information Science and Artificial Intelligence*, 1350–1356. <https://doi.org/10.1145/3773365.3773578>
- [9] Di Guardo, A., Trovato, F., Cantisani, C., Dattola, A., Nisticò, S. P., Pellacani, G., & Paganelli, A. (2025). Artificial intelligence in cosmetic formulation: Predictive modeling for safety, tolerability, and regulatory perspectives. *Cosmetics*, 12(4), 157. <https://doi.org/10.3390/cosmetics12040157>
- [10] Liu, Y., Liu, Y., Zhang, S., Zeng, C., Zhang, Q., Jiang, Y., Yang, X., Zheng, L., Ge, Q., Zhang, Y., Chen, Y., Lu, M., & Liu, H. (2025). Using explainable machine learning to predict the irritation and corrosivity of chemicals on eyes and skin. *Toxicology Letters*, 408, 1–12. <https://doi.org/10.1016/j.toxlet.2025.03.008>
- [11] Lechuga, M., García, P. A., García-López, A. I., Tapia-Navarro, C., & Ríos, F. (2025). Development of a predictive classification model for surfactant-induced skin irritation. *ACS Omega*, 10(46), 55868–55878. <https://doi.org/10.1021/acsomega.5c07338>
- [12] Xie, J., Liu, W., Hu, W., Ouyang, M., & Huang, T. (2025). Graph neural network-based toxicity prediction by integrating molecular fingerprints and knowledge graph features. *Toxics*, 13(11), 953. <https://doi.org/10.3390/toxics13110953>

- [13] Zhang, K., Schwaller, P., & Fenner, K. (2025). Predicting toxicity toward nitrifiers by attention-enhanced graph neural networks and transfer learning from baseline toxicity. *Environmental Science & Technology*, 59(9), 4518–4529. <https://doi.org/10.1021/acs.est.4c12247>
- [14] Lee, J., Yoon, H., Kim, S., Lee, C., Lee, J., & Yoo, S. (2024). Deep learning-based skin care product recommendation: A focus on cosmetic ingredient analysis and facial skin conditions. *Journal of Cosmetic Dermatology*, 23(6), 2066–2077. <https://doi.org/10.1111/jocd.16218>
- [15] Xin, H., Virk, A. S., Virk, S. S., Akin-Ige, F., & Amin, S. (2024). Applications of artificial intelligence and machine learning on critical materials used in cosmetics and personal care formulation design. *Current Opinion in Colloid & Interface Science*, 73, 101847. <https://doi.org/10.1016/j.cocis.2024.101847>
- [16] Grech, V. S., Kefala, V., & Rallis, E. (2024). Cosmetology in the era of artificial intelligence. *Cosmetics*, 11(4), 135. <https://doi.org/10.3390/cosmetics11040135>
- [17] Huang, Z., Lou, S., Wang, H., Li, W., Liu, G., & Tang, Y. (2024). AttentiveSkin: To predict skin corrosion/irritation potentials of chemicals via explainable machine learning methods. *Chemical Research in Toxicology*, 37(2), 361–373. <https://doi.org/10.1021/acs.chemrestox.3c00332>
- [18] Wang, H., Huang, Z., Lou, S., & Tang, Y. (2024). In silico prediction of skin sensitization for compounds via flexible evidence combination based on machine learning and Dempster–Shafer theory. *Chemical Research in Toxicology*, 37(6), 894–909. <https://doi.org/10.1021/acs.chemrestox.3c00396>
- [19] Qiao, W., Xie, T., Lu, J., & Jia, T. (2024). Development of machine learning models for the prediction of the skin sensitization potential of cosmetic compounds. *PeerJ*, 12, e18672. <https://doi.org/10.7717/peerj.18672>
- [20] Kong, W., Zhu, J., Shan, P., & Ying, H. (2024). iSKIN: Integrated application of machine learning and Mondrian conformal prediction to detect skin sensitizers in cosmetic raw materials. *SmartMat*, 5(6), e1278. <https://doi.org/10.1002/smm2.1278>